

Subnetting homework

2024

- Assigned network: 192.168.24.0
- Host address requests:
 - 16 addresses for range A
 - 127 addresses for range B
 - 30 addresses for range C
 - 15 addresses for range D
 - 63 addresses for range E
 - 7 addresses for range F
- Task: ordering requests from bigger to lower, and assigning IP addresses from lower to higher, divide the original set to satisfy all requests.

1. Ajustar

rang a = 32 (2^5)
 range b = 256 (2^8)
 range c = 32 (2^5)
 range d = 32 (2^5)
 range e = 128 (2^7)
 range f = 16 (2^4)

256

2^9 address necessary



mask: $32 - 9 = 23$

2. IP: 192.168.24.0 / 23 ^{max 255}

Bigger set with higher ID.

Net + N - 1

Net + 1

BC - 1

Set	N	Net	Mask	BC	First	Last
A	32	(9) 192.168.24.32	27	192.168.24.63	192.168.24.33	192.168.24.62
B	256	(1) 192.168.25.0	24	192.168.25.255	192.168.25.1	192.168.25.254
C	32	(4) 192.168.24.64	27	192.168.24.95	192.168.24.65	192.168.24.94
D	32	(3) 192.168.24.96	27	192.168.24.127	192.168.24.97	192.168.24.126
E	128	(2) 192.168.24.128	25	192.168.24.255	192.168.24.129	192.168.24.254
F	16	(6) 192.168.24.16	28	192.168.24.31	192.168.24.17	192.168.24.30

(1) 192.168.24.0

+ 256 = 192.168.25.0

- 128 = 192.168.24.128

(d) - 32 = 192.168.24.96

(c) - 32 = 192.168.24.64

(a) - 32 = 192.168.24.32

- 16 = 192.168.24.16

Bigger set with higher ID.

 $Net + N - 1$

$Net + 1$

$BC - 1$

Set	N	Net	Mask	BC	First	Last
A	32	(3) 192.168.25.128	27	192.168.25.159	192.168.25.129	192.168.25.158
B	256	(1) 192.168.24.0	24	192.168.24.255	192.168.24.1	192.168.24.254
C	32	(4) 192.168.25.160	27	192.168.25.191	192.168.25.161	192.168.25.190
D	32	(5) 192.168.25.192	27	192.168.25.223	192.168.25.193	192.168.25.222
E	128	(2) 192.168.25.0	25	192.168.25.127	192.168.25.1	192.168.25.126
F	16	(6) 192.168.25.224	28	192.168.25.239	192.168.25.225	192.168.25.238